Applications of Lasers, Plasma and Accelerators in Agriculture and Medicine

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- Accelerating India towards Self Reliance
  - Technology
  - Creation of More Wealth
  - Higher Standards of Living

- Developing Lasers, Accelerators, Plasma, Supercomputers, Instrumentation

- Radiation Technology Research
  - Food Security
  - Treatment of deadly diseases
  - Environmental protection
  - Industrial development

- Supporting Fundamental Research [BARC, RRCAT, IPR, VECC, IoP, NISER, HRI, AMD]

Fig: Map of DAE Research Centres, PSU's, Industrial Organisations and Service Centres across India

LASER, Plasma and Accelerator

DAE Contribution

- Lasers for industry and medicine (RRCAT)
- DAE ILU-6 at BRIT : sterilization, imaging, diseases, quality enhancement
- FCIPT: Plasma Steriliser, Anti Microbial Coating for Corona, Plasma activated water

Fig: Portable Hand Sterilizer

Fig: 20kW LASER, RRCAT

- First **Semiconductor Laser** Developed at BARC 1965
- BARC-TIFR **Optical Communication** link 1966
- **1 MV Cockroft Walton** (TIFR, Bombay) and Low Energy Cyclotron at SINP
- Contribution to ITER: **Cryostat**, **ICRF** source systems

Importance

Lasers: high precision operations, cutting, fiber optics, meteorology, delicate surgery

Plasma: Sterilization, Welding, Displays, Waste to Energy, Blood Coagulation, Killing harmful cell, pesticide removal

Accelerators: Isotope generation, sterilization, food preservation, ion implantation, ion beam processing

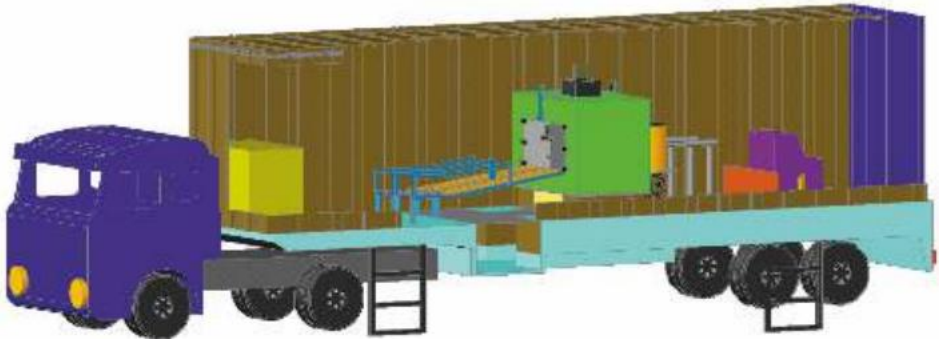
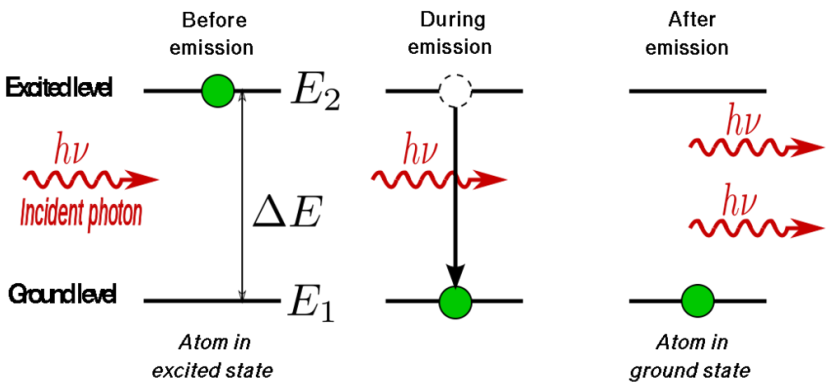


Fig: Proposed Mobile Food Irradiator(MFI), BRIT

Lasers

- **L**ight **A**mplification by **S**timulated **E**mission of **R**adiation

- Produces High Intensity
  1.) **Coherent**
  2.) **Collimated**
  3.) **Monochromatic**



$$E_2 - E_1 = \Delta E = h\nu$$

Fig: Mechanism of Laser – Population Inversion and Stimulated Emission

Laser Types
Gas Laser
Solid State Laser
Semiconductor Laser
Free Electron Laser
Liquid Lasers (Dye Lasers)

Components of LASER

1.) Gain Medium: medium for **population inversion.**
2.) Pump Source: Energy source for inversion.
3.) Resonator: Feedback for amplification.

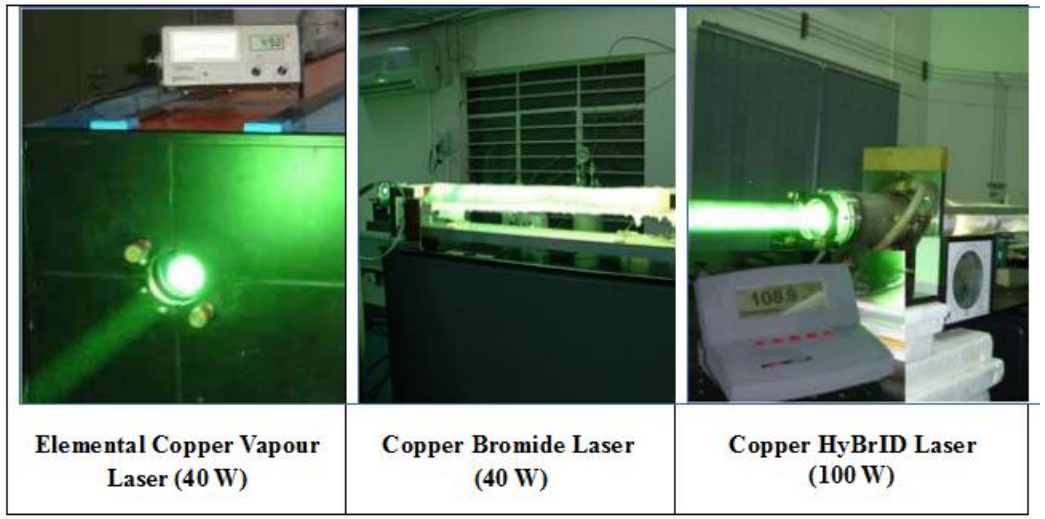


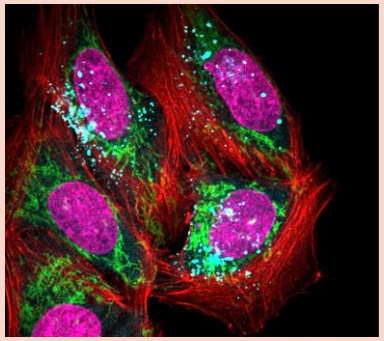
Fig: Copper Vapour Laser developed at RRCAT

Application of Lasers

Medicine

1.) Surgery(Bloodless Cuts, Unwanted Tissue Removal[**CO2**]
, Retina Fixing, Cataract) [**Nd-YAG**]
2.) Transmitted to remotest part of body(kidney stones)
3.) Balloon Angioplasty: Blocked Arteries
4.) Arthroscopic Surgery, Cosmetics, Skin Complications
5.) Confocal Microscopy

LASIK surgery



Confocal Image

Agriculture

1.) On Farm Navigation Systems (High Precision)
2.) Land Levelling, Saves Water
3.) Assessing Fertiliser Purity, Grain Yield and Size, Fertility(Scanning soil Density)
4.) Enhancing Plant Growth, Killing weeds
5.) Disease Resistant Crops (Laser Marking Machines)
6.) Laser Scare Crows

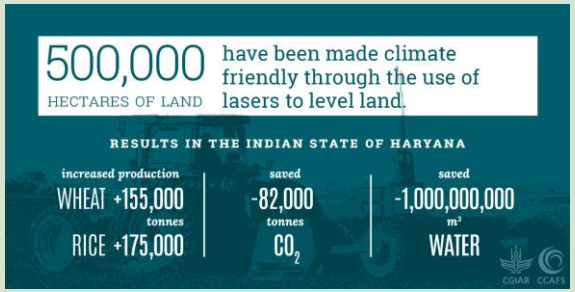


Fig: Statistics on land revival and water saved

Fig: Lase based land levelling system

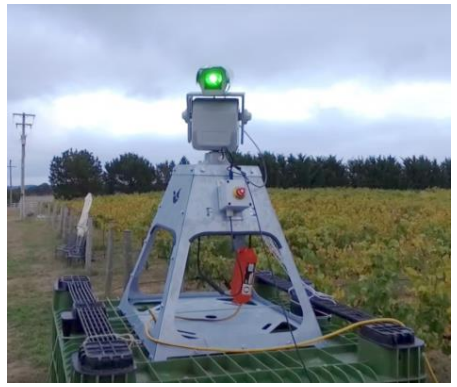


Fig: Laser Scarecrow

Fig: LASER Scanners for soil density

Plasma

- Super Heated Matter
- Electrons ripped away from the atoms forming an ionized gas.
- Most common state of matter
- Equal number of ionised atoms and free electrons

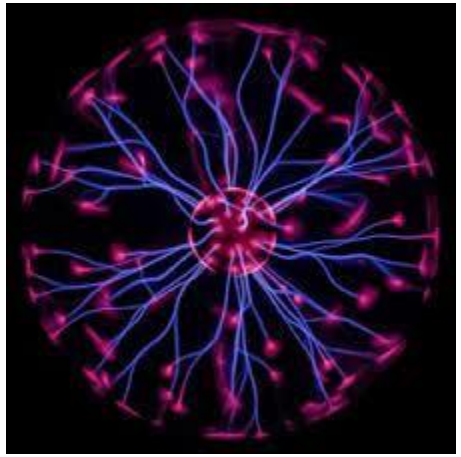


Fig: Plasma-*Forth State of Matter*

Types of Plasmsa

Temperature	Degree of Ionisation	Density
1.) Very Hot 2.) Hot 3.) Cold 4.) Very Cold	1.) Fully Ionised 2.) Partially Ionised	1.)High Density 2.) Medium Density 3.) Low Density

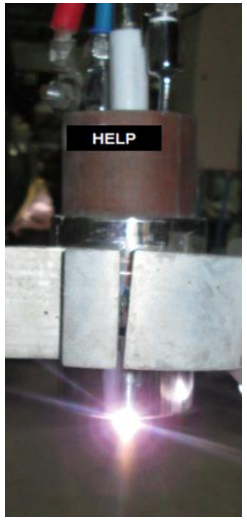
Other Types: Magnetic, Non Magnetic; dusty, grain, colloidal, ideal non ideal plasma, high energy density plasma

Properties

- No definite shape/size
- Electrically neutral
- Strong Interaction with EM fields
- High temperature, high enthalpy and energy density
- Steep temperature gradient

Generation of Plasma

- Electrical Arc
- DC Glow Discharge (Helicon)
- Capacitively Coupled Plasma
- Cascaded Arc Plasma Source
- Electron Cyclotron Resonance
- Dielectric Barrier Discharge



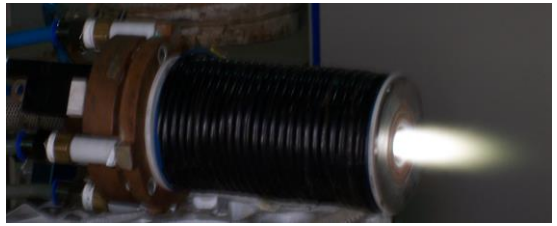


Fig: Plasma Torches developed at BARC

Applications of Plasma

Medicine

1.) Plasma Medicine
2.) Sterilization (Dental Implants, Surgical Equipment)
3.) Modify Biomaterial Properties – making useful surfaces.
4.) Cure skin diseases, wounds, disinfection of skin surface (in vitrio) [DBD]
5.) Killing harmful cells, bacterial inactivation, Treatment of Viable human tissues (Low Temperature Plasma Jet)

Cold Plasma Sterilization System

Agriculture

1.) Plasma Activated Water
- Prevent bacterial growth in milk
- Increased rooting speed, reduce water consumption
- Enhance Seed Germination
- Prevent Pests

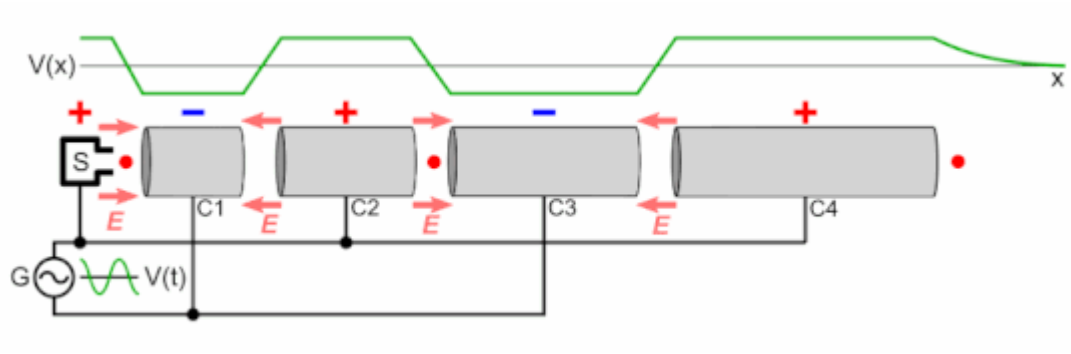
2.) Seed Sterilization @ Storage Facility
3.) Air cleaning, Sterilization, VOC's removal in greenhouse plants
4.) Soil Remediation
5.) Waste to Energy conversion of non hazardous waste.
6.) Reducing foodstuff ageing by removing ethylene ($H_2C{=}CH_2$).

Air Plasma Gasifier Unit for Waste Management

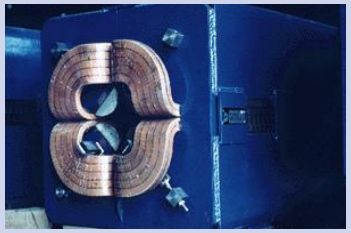
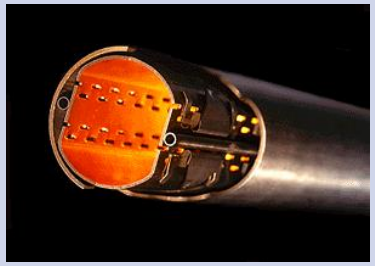
Plasma treatment of food using atmospheric air

Accelerators

- Accelerate fundamental particles to very high energies
- Use Oscillating electric field
- Magnetic fields => Focus Particles
- Fixed Target or Two Beam Targets



Types
Linear
Cyclotron
Synchrotron

Accelerating Component	Bending Component	Focusing Component	Race Track
Cavity(Klystron)	Dipole Magnet	Quadrupole/Sextupole Magnet 	Beam Pipe Copper(Vacuum Chamber) 

ADVANTAGES

- Electrical devices: ON/OFF
- No radioactive material
- No recontamination
- Zero chemical residues
- Maintains Aroma
- Time Efficient than Gamma
- Proton/Ion therapy

Applications of Accelerators

Medicine

1.) Radio Isotope Production

Tumor Imaging 18-F
Cancer Imaging Ga-67
Lung Disorders 81m-Kr
Blood Flow Rate 123-I
Myocardial Scintigraphy 201-Tl
Antibodies Labelling 111-In
Medical imaging

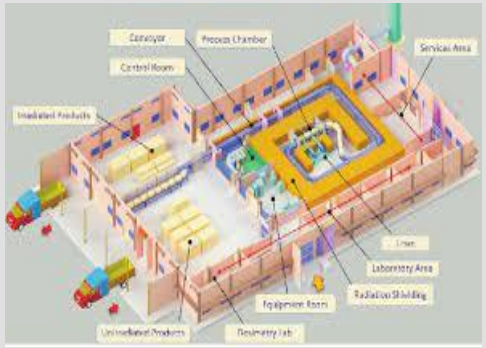
Fig: Clinac-2100C at Tata Memorial Hospital

2.) X Ray Radiotherapy: Cancer Treatment
3.) Proton Electron Radiotherapy: curing childhood cancers
Microtron Accelerator at RRCAT
4.) Sterilization of Medical Disposables (Electron Accelerators)

Agriculture

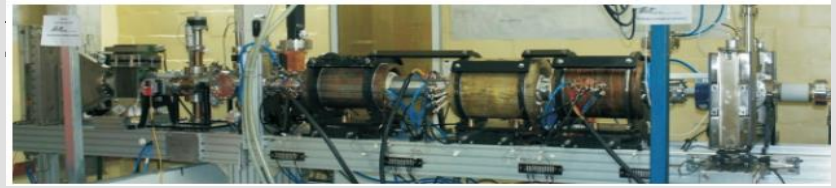
1.) Food Irradiation
- Increased Shelf Life
- Production Enhancement
- Reducing Microbial Growth

Agriculture Radiation Processing Facility, RRCAT (10 MeV e Beam)



ARPF

2.) Phytosanitary Treatment of agriculture produce for export
3.) Microbial Decontamination of Spices and Ayurveda Products
4.) Mutation Breeding and Improved seed qualities (Seed Irradiation).
5.) Hydrogel, Chitosan (bio stimulator) for better crop



Indus II at RRCAT

Conclusions

- Lasers Plasma and Accelerator Technology play a crucial role in two most important aspects of humankind, food and medicine.
- DAE played a crucial role in not only in development of these technologies but also ensuring that benefits of these are reaped till the **last mile**.
- DAE with its **multidimensional activities** and **units** has placed India in top position in the field of innovations in medicine and agriculture.
- India's contribution in mega science projects like ITER, CERN, LIGO, INO and SKA have been commendable. It will surely go a long way in India's expedition towards scientific and technological supremacy.

India's Contribution in Mega Science Projects

Future Outlook

- Lasers Plasma and Accelerator technology offer great potential solutions to innumerable problems in future.

- In Future, lasers may be used for quantum communication and measurement of gravitational waves. They will become smaller and more powerful, into quantum realm.

- Accelerators will render 3D images and movies of electron collisions with protons/ions. They will be coupled with Machine Learning, Lasers and Superconduction to reinvent themselves. CT Scanner for atoms ?

- Plasma Technology will aid in shifting from conventional sources of energy to electricity in industrial processes. Discovering newer states of matter, gaining insights into mysterious universe processes, improvements in microelectronics, biotechnologies and water purification are anticipated.

Image Credits

1.) DAE, BARC, RRCAT, IPR, BRIT websites, publications.

2.)http://www.madrimasd.org/cienciaysociedad/ateneo/dossier/particulas/cern/public_web_cern_ch/Public/accelerators/components.html

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4.) https://vigyansamagam.in/mega-science-projects [Logos of Mega Science Projects]

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6.) https://mapio.net/images-p/48026006.jpg [Last Page Image Background]

7.) Special Issues on Plasma in Agriculture and Food Cycle https://cms.iopscience.org

8.) https://www.barc.gov.in/pubaware/agri.html [Background Image on First Slide]

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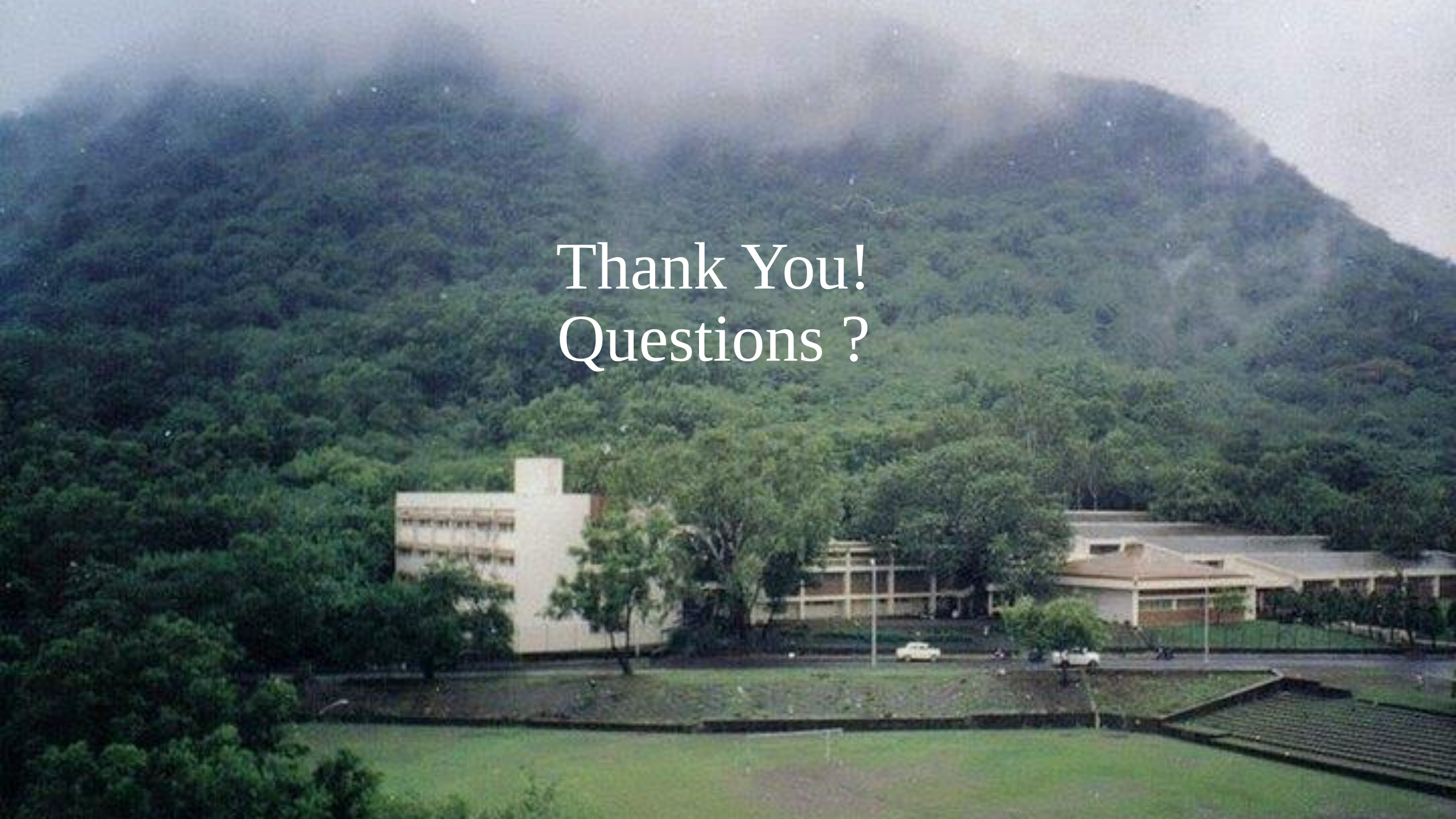

Thank You!
Questions ?